LAYUP ANALYSIS TOOL FOR MATLAB®

Analytical Strength Assessment and Layup Optimisation Code for Unidirectional Fibre-reinforced Composites

Analysis User's Guide



Acknowledgements

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1. Strength Theory Reference Guide

1.1 Failure and damage initiation criteria for fibre-reinforced composites

Composite failure criteria are used to assess the strength of fibre-reinforced composite laminate components subject to biaxial loading.

A composite can fail in two basic modes:

- Failure of individual plies in tension, compression, or shear (fibre and matrix); and
- delamination between plies.

1.2 About failure and damage initiation criteria for fibre-reinforced composites

These composite failure modes are illustrated by Figure 1.2.1. Layup Analysis Tool is able to assess the first mode of failure.

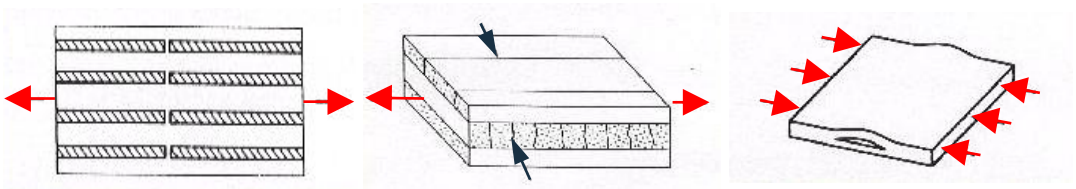


Figure 1.2.1: (L-R) Fibre, matrix and delamination failure

1.3 Conventions for fibre-reinforced composites

The composite failure criteria assessment assumes that the loading is in a state of orthotropic plane stress. The ply directions are given by Equation 1.3.1.

$$\boldsymbol{\sigma} = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{bmatrix} \equiv \begin{bmatrix} \sigma_{fibre} & \sigma_{shear} & 0 \\ \sigma_{shear} & \sigma_{transverse} & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad [1.3.1]$$

Layup Analysis Tool requires stress components σ_{11}, σ_{22} and σ_{12} for the fibre, transverse and shear directions, respectively.

1.4 Material properties

1.4.1 Stress-based failure criteria

MATLAB property ID	Property name	Symbol
XT	Tensile stress limit in fibre direction	X_T
XC	Compressive stress limit in fibre direction	X_C
YT	Tensile stress limit in transverse direction	Y_T
YC	Compressive stress limit in transverse direction	Y_C
S	Shear strength in the 12-plane	S
C	Cross-product term coefficient in the 12-plane	f^*_{12}
B	Biaxial stress limit in the 12-plane	$\sigma_{biax,12}$

1.4.2 Maximum strain failure theory

MATLAB property ID	Property name	Symbol
XET	Tensile strain limit in fibre direction	$X_{\varepsilon T}$
XEC	Compressive strain limit in fibre direction	$X_{\varepsilon C}$
YET	Tensile strain limit in transverse direction	$Y_{\varepsilon T}$
YEC	Compressive strain limit in transverse direction	$Y_{\varepsilon C}$
SE	Shear strain limit in the 12-plane	S_{ε}

1.4.3 Hashin's damage initiation theory

MATLAB property ID	Property name	Symbol
ALPHA	Shear influence parameter	α
XHT	Longitudinal tensile strength of the lamina	X_{hT}
XHC	Longitudinal compressive strength of the lamina	X_{hC}
YHT	Transverse tensile strength of the lamina	Y_{hT}
YHC	Transverse compressive strength of the lamina	Y_{hC}
SHX	In-plane shear strength of the lamina	S_{hX}
SHY	Transverse shear strength of the lamina	S_{hY}

1.4.4 LaRC05 damage initiation theory

MATLAB property ID	Property name	Symbol
XLT	Longitudinal tensile strength	X_{lT}
XLC	Longitudinal compressive strength	X_{lC}
YLT	Transverse tensile strength	Y_{lT}
YLC	Transverse compressive strength	Y_{lC}
SLX	In-plane shear strength	S_{lX}
SLY	Transverse shear strength	S_{lY}
GL12	In-plane shear modulus in the 12-plane	$G_{l,12}$
NL	Longitudinal shear friction coefficient	η_L
NT	Transverse shear friction coefficient	η_T
A0	Fracture plane angle for pure compression	α_0
PHI0	Misalignment angle at failure for pure compression	φ^0

1.5 Maximum stress theory

The maximum stress theory predicts composite failure by comparing each stress component to its respective stress limit, according to Equation 1.5.1.

$$I_F = max\left(\frac{\sigma_{11}}{X}, \frac{\sigma_{22}}{Y}, \left|\frac{\sigma_{12}}{S}\right|\right) < 1 \quad [1.5.1]$$

The value of the tensile (fibre) stress limit, X, and transverse stress limit, Y, is determined according to Equations 1.5.2-3.

$$\sigma_{11} > 0 \rightarrow X = X_T;\ \sigma_{11} \leq 0 \rightarrow X = X_C \quad [1.5.2]$$

$$\sigma_{22} > 0 \rightarrow Y = Y_T;\ \sigma_{22} \leq 0 \rightarrow Y = Y_C \quad [1.5.3]$$

This theory does not account for the interaction between the stress components in different directions. Therefore, its accuracy is limited.

1.6 Tsai-Hill theory

Tsai proposed a composite failure criterion as an extension of Hill's anisotropic plasticity model [1]. This simple model defines the failure criterion as a function of the first three Cauchy (plane stress) tensor components according to Equation 1.6.1.

$$I_F = max\left(\frac{{\sigma_{11}}^2}{X^2} - \frac{\sigma_{11}\sigma_{22}}{X^2} + \frac{{\sigma_{22}}^2}{Y^2} + \frac{{\sigma_{12}}^2}{S^2}\right) < 1 \qquad [1.6.1]$$

The value of the tensile (fibre) stress limit, X, and transverse stress limit, Y, is determined according to Equations 1.5.2-3.

The Tsai-Hill theory is a piecewise-continuous failure surface, shown by Figure 1.6.1.

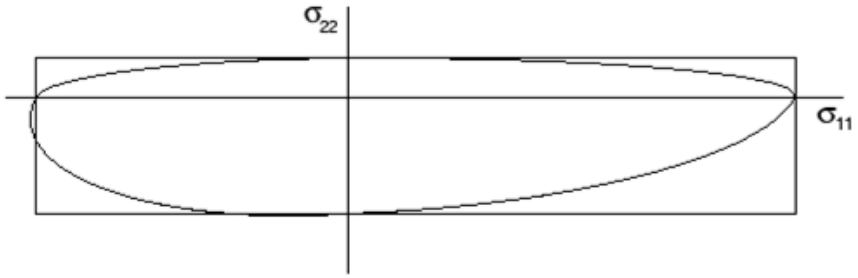


Figure 1.6.1: Tsai-Hill piecewise-continuous failure surface

1.7 Hoffman theory

The Hoffman failure criterion is a triaxial version of the Tsai-Hill failure criterion [2]. It is based on a quadratic interaction between the stress components. The Hoffman failure criterion has been used in the design of high-stress composite structures such as aerospace components where detailed failure prediction is crucial.

This model defines the failure criterion as a function of the six Cauchy stress tensor components according to Equation 1.7.1.

$$I_F = C_1(\sigma_{22} - \sigma_{33})^2 + C_2(\sigma_{33} - \sigma_{11})^2 + C_3(\sigma_{11} - \sigma_{22})^2 + C_4\sigma_{11} + C_5\sigma_{22} + C_6\sigma_{33} + C_7{\sigma_{23}}^2 + C_8{\sigma_{13}}^2 + C_9{\sigma_{12}}^2 < 1 \quad [1.7.1]$$

where C_1-C_9 are material parameters given by Equations 1.7.2-6.

$$C_1 = \frac{1}{2}\left[\frac{1}{X_C X_T} - \frac{1}{Y_C Y_T} - \frac{1}{Z_C Z_T}\right] \quad [1.7.2]$$

$$C_2 = \frac{1}{2}\left[-\frac{1}{X_C X_T} + \frac{1}{Y_C Y_T} - \frac{1}{Z_C Z_T}\right] \quad [1.7.3]$$

$$C_3 = \frac{1}{2}\left[-\frac{1}{X_C X_T} - \frac{1}{Y_C Y_T} + \frac{1}{Z_C Z_T}\right] \quad [1.7.4]$$

$$C_4 = \frac{1}{X_T} + \frac{1}{X_C},\ C_5 = \frac{1}{Y_T} + \frac{1}{Y_C},\ C_6 = \frac{1}{Z_T} + \frac{1}{Z_C} \quad [1.7.5]$$

$$C_7 = C_8 = C_9 = \frac{1}{S^2} \quad [1.7.6]$$

Note: Layup Analysis Tool assumes a state of two-dimensional plane stress. Therefore, the following assumptions are made: $Z_T = X_T$, $Z_C = X_C$ and $\sigma_{33} = \sigma_{23} = \sigma_{12} = 0$.

1.8 Tsai-Wu theory for fibre-reinforced composites

The Tsai-Wu composite failure criterion is a smooth form of the Tsai-Hill criterion [3]. The direct and shear stress is coupled by the parameter F_{12}, which is a function of either the cross-product coefficient, f^*_{12} or the biaxial stress limit, $\sigma_{biax,12}$.

The criterion is defined according to Equation 1.8.1.

$$I_F = F_1\sigma_{11} + F_2\sigma_{22} + F_{11}{\sigma_{11}}^2 + F_{22}{\sigma_{22}}^2 + F_{66}{\sigma_{12}}^2 + 2F_{12}\sigma_{11}\sigma_{22} < 1 \quad [1.8.1]$$

The Tsai-Wu coefficients are defined according to Equation 1.7.2.

$$F_1 = \frac{1}{X_T} + \frac{1}{X_C}, F_2 = \frac{1}{Y_T} + \frac{1}{Y_C}, F_{11} = -\frac{1}{X_T X_C}, F_{22} = -\frac{1}{Y_T Y_C}, F_{66} = \frac{1}{S^2} \quad [1.8.2]$$

If the biaxial stress limit, $\sigma_{biax,12}$, is defined, the coupling parameter, F_{12}, is given by Equation 1.8.3.

$$F_{12} = \frac{1}{2{\sigma_{biax,12}}^2}\left[1 - \left(\frac{1}{X_T} + \frac{1}{X_C} + \frac{1}{Y_T} + \frac{1}{Y_C}\right)\sigma_{biax,12} + \left(\frac{1}{X_T X_C} + \frac{1}{Y_T Y_C}\right){\sigma_{biax,12}}^2\right] \quad [1.8.3]$$

If the biaxial stress limit is not defined, the coupling parameter is defined by Equation 1.8.4.

$$F_{12} = f^*_{12}\sqrt{F_{11}F_{22}}, \{-1 \le f^*_{12} \le 1\} \quad [1.8.4]$$

The default value of f^*_{12} is zero. If $F_{12} = 0$, the Tsai-Wu failure surface is given by Figure 1.8.1.

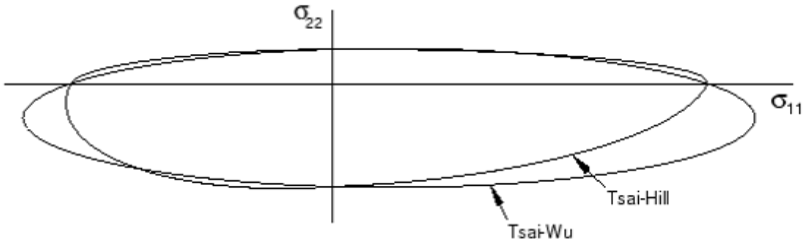


Figure 1.7.1: Tsai-Wu piecewise-continuous failure surface ($F_{12} = 0$)

1.9 Azzi-Tsai-Hill theory

The Azzi-Tsai-Hill composite failure criterion is the same as the Tsai-Hill theory, except that the absolute value of the cross-product term coefficient is used, according to Equation 1.9.1.

$$I_F = max\left(\frac{{\sigma_{11}}^2}{X^2} - \left|\frac{\sigma_{11}\sigma_{22}}{X^2}\right| + \frac{{\sigma_{22}}^2}{Y^2} + \frac{{\sigma_{12}}^2}{S^2}\right) < 1 \qquad [1.9.1]$$

The Azzi-Tsai-Hill failure criterion has been used in the design of composite pressure vessels where quick and reasonably accurate failure predictions are required.

The Azzi-Tsai-Hill failure surface is given by Figure 1.9.1.

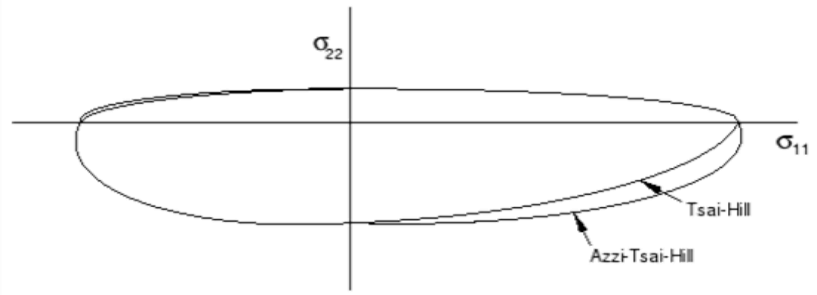


Figure 1.9.1: Azzi-Tsai-Hill piecewise-continuous failure surface

1.10 Maximum strain failure theory

The maximum failure strain theory predicts composite failure by comparing each strain component to its respective strain limit, according to Equation 1.10.1.

$$I_F = max\left(\frac{\varepsilon_{11}}{X_\varepsilon}, \frac{\varepsilon_{22}}{Y_\varepsilon}, \left|\frac{\gamma_{12}}{S_\varepsilon}\right|\right) < 1 \quad [1.10.1]$$

The value of the tensile (fibre) strain limit, X_ε, and transverse strain limit, Y_ε, is determined according to Equations 1.10.2-3.

$$\varepsilon_{11} > 0 \rightarrow X_\varepsilon = X_{\varepsilon T};\ \varepsilon_{11} \leq 0 \rightarrow X_\varepsilon = X_{\varepsilon C} \quad [1.10.2]$$

$$\varepsilon_{22} > 0 \rightarrow Y_\varepsilon = Y_{\varepsilon T};\ \varepsilon_{22} \leq 0 \rightarrow Y_\varepsilon = Y_{\varepsilon C} \quad [1.10.3]$$

Experimental data suggests that the maximum failure strain theory is not as accurate as the maximum stress theory.

The maximum failure strain theory assumes that the stresses in the composite are linear elastic until $I_F = 1$. The elastic stress is converted into elastic strain using the shear modulus, $G_{\varepsilon,12}$, according to Equation 1.10.4.

$$\varepsilon_{11} = \frac{\sigma_{11}}{E_{11}}, \varepsilon_{22} = \frac{\sigma_{22}}{E_{22}}, \gamma_{12} = \frac{\sigma_{12}}{G_{\varepsilon,12}} \quad [1.10.4]$$

1.11 Hashin's damage initiation criteria

The damage initiation criteria for fibre-reinforced composites are calculated using Hashin's theory [4]. The model considers the following modes of damage initiation:

I. Fibre tension ($I_{D,ft}$)
II. Fibre compression ($I_{D,fc}$)
III. Matrix tension ($I_{D,mt}$)
IV. Matrix compression ($I_{D,mc}$)

Hashin's theory has been applied to the design of composite automotive components to ensure reliability under different loading scenarios.

The criteria for each mode are calculated according to Equations 1.11.1-4.

$$I_{D,ft} = \left(\frac{\widehat{\boldsymbol{\sigma}}_{11}}{X_{hT}}\right)^2 + \alpha\left(\frac{\widehat{\boldsymbol{\sigma}}_{12}}{S_{hL}}\right)^2 = 1, \{0 \le \alpha \le 1;\ \widehat{\boldsymbol{\sigma}}_{11} \ge 0\} \quad [1.11.1]$$

$$I_{D,fc} = \left(\frac{\widehat{\boldsymbol{\sigma}}_{11}}{X_{hC}}\right)^2 = 1, \{\widehat{\boldsymbol{\sigma}}_{11} < 0\} \quad [1.11.2]$$

$$I_{D,mt} = \left(\frac{\widehat{\boldsymbol{\sigma}}_{22}}{Y_{hT}}\right)^2 + \left(\frac{\widehat{\boldsymbol{\sigma}}_{12}}{S_{hL}}\right)^2 = 1, \{\widehat{\boldsymbol{\sigma}}_{22} \ge 0\} \quad [1.11.3]$$

$$I_{D,mc} = \left(\frac{\widehat{\boldsymbol{\sigma}}_{22}}{2S_{hT}}\right)^2 + \left[\left(\frac{Y_C}{2S_{hT}}\right)^2 - 1\right]\frac{\widehat{\boldsymbol{\sigma}}_{22}}{Y_{hC}} + \left(\frac{\widehat{\boldsymbol{\sigma}}_{12}}{S_{hL}}\right)^2 = 1, \{\widehat{\boldsymbol{\sigma}}_{22} < 0\} \quad [1.11.4]$$

The effective stress tensor, $\widehat{\boldsymbol{\sigma}}$, is given by Equation 1.11.5.

$$\widehat{\boldsymbol{\sigma}} \equiv \begin{bmatrix} \sigma_{11} & 0 & 0 \\ 0 & \sigma_{22} & 0 \\ 0 & 0 & \sigma_{12} \end{bmatrix} \quad [1.11.5]$$

The shear influence parameter, α, is the coefficient that multiplies the shear contribution to Hashin's fibre initiation criterion. The default value is $\alpha = 0.0$.

1.12 LaRC05 damage initiation criteria

Overview

An alternative damage initiation criterion has been proposed by Pinho et al. [5] [6], wherein the damage in the composite is based on a pressure-dependent, three-dimensional constitutive law. The model considers the following modes of failure:

I. Polymer ($I_{F,p}$)
II. Matrix ($I_{F,m}$)
III. Fibre kinking/splitting ($I_{F,k/s}$)
IV. Fibre tensile ($I_{F,t}$)

The LaRC05 criterion has been used to make predictions of the test cases presented from the second World-Wide Failure Exercise [7]. It has also been applied to the design of space structures and high-performance aircraft components, where precise failure prediction is essential.

Polymer failure

The polymer damage initiation criterion is calculated according to Equation 1.12.1.

$$I_{D,p} = \frac{3[k^2 - (X_{lT} - X_{lC})\sigma_h]}{X_{lT}X_{lC}} \quad [1.12.1]$$

The hydrostatic stress is given by Equation 1.12.2.

$$3\sigma_h = \sigma_{11} + \sigma_{22} + \sigma_{33} \quad [1.12.2]$$

The parameter, k, is given by Equation 1.12.3.

$$6k^2 = (\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2 \quad [1.12.3]$$

The maximum, middle and minimum principal stresses are given by σ_1, σ_2 and σ_3, respectively.

Matrix failure

The matrix damage initiation criterion is calculated according to Equation 1.12.4.

$$I_{D,m} = \sqrt{\left(\frac{\tau_T}{S_{lY} - \eta_T\sigma_N}\right)^2 + \left(\frac{\tau_L}{S_{lX} - \eta_L\sigma_N}\right)^2 + \left(\frac{\langle\sigma_N\rangle_+}{Y_{lT}}\right)^2} \quad [1.12.4]$$

The stresses τ_T, τ_L and σ_N are the traction components on each trial fracture plane, and are calculated according to Equations 1.12.5-7.

$$\sigma_N = \frac{\sigma_{22} + \sigma_{33}}{2} + \frac{\sigma_{22} + \sigma_{33}}{2}\cos(2\alpha) + \sigma_{23}\sin(2\alpha) \quad [1.12.5]$$

$$\tau_T = -\frac{\sigma_{22} - \sigma_{33}}{2}\sin(2\alpha) + \sigma_{23}\cos(2\alpha) \quad [1.12.6]$$

$$\tau_L = \sigma_{12}\cos(\alpha) + \sigma_{13}\sin(\alpha) \quad [1.12.7]$$

Fibre kinking/splitting failure

The fibre kinking and splitting damage initiation criteria are calculated according to Equation 1.12.8.

$$I_{D,k/s} = \sqrt{\left(\frac{\tau^m{}_{23}}{S_{lY} - \eta_T \sigma^m{}_2}\right)^2 + \left(\frac{\tau^m{}_{12}}{S_{lY} - \eta_L \sigma^m{}_2}\right)^2 + \left(\frac{\langle \sigma^m{}_2 \rangle_+}{Y_{lT}}\right)^2} \quad [1.12.8]$$

The stresses on the kink band plane are given by Equations 1.12.9-12.

$$\sigma^\psi{}_{22} = \sigma_{22} \cdot \cos(\psi)^2 + \sigma_{33} \cdot \sin(\psi)^2 + 2\tau_{23} \cdot \sin(\psi)\cos(\psi) \quad [1.12.9]$$

$$\tau^\psi{}_{12} = \tau_{12}\cos(\psi) + \tau_{31}\sin(\psi) \quad [1.12.10]$$

$$\tau^\psi{}_{23} = \left(\sigma^\psi{}_{22} - \sigma_{11}\right)\sin\psi\cos\psi + \tau_{23}[\cos(\psi)^2 - \sin(\psi)^2] \quad [1.12.11]$$

$$\tau^\psi{}_{31} = \tau_{31}\cos(\psi) - \tau_{12}\sin(\psi) \quad [1.12.12]$$

The stresses on the fibre misalignment frame are given by Equations 1.12.13-15.

$$\sigma^m{}_2 = \sigma_{11} \cdot \sin(\varphi)^2 + \sigma^\psi{}_{22} \cdot \cos(\varphi)^2 - 2\tau^\psi{}_{12} \cdot \sin(\varphi)\cos(\varphi) \quad [1.12.13]$$

$$\tau^m{}_{12} = \left(\sigma^\psi{}_{22} - \sigma_{11}\right)\sin\varphi\cos\varphi + \tau^\psi{}_{12}[\cos(\varphi)^2 - \sin(\varphi)^2] \quad [1.12.14]$$

$$\tau^m{}_{23} = \tau^\psi{}_{23}\cos(\varphi) - \tau^\psi{}_{31}\sin(\varphi) \quad [1.12.15]$$

The misalignment angle, φ, is the sum of the initial misalignment angle, φ^0, and the shear strain in the initial fibre misalignment frame, γ_{m^0}, and is given by Equation 1.12.16.

$$\varphi = s(\varphi^0 + \gamma_{m^0}), \qquad s = \begin{cases} -1, & \tau^\psi{}_{12} < 0 \\ +1, & \tau^\psi{}_{12} \geq 0 \end{cases}$$

$$\gamma_{m^0} = \frac{1}{G_{12}}\left|\frac{\sigma^\psi{}_{22} - \sigma_{11}}{2}\sin(2\varphi^0) + \left|\tau^\psi{}_{12}\right|\cos(2\varphi^0)\right| \quad [1.12.16]$$

Fibre tensile failure

The tensile damage initiation criterion is calculated according to Equation 1.12.17.

$$I_{D,t} = \frac{\langle \sigma_{11} \rangle_+}{X_{lT}} \quad [1.12.17]$$

Derived properties

If the value of S_{lY} is not specified, it is derived from Equation 1.12.18.

$$S_{lY} = Y_{lC} \cos(\alpha_0) \left[\sin(\alpha_0) + \frac{\cos(\alpha_0)}{\tan(2\alpha_0)} \right] \quad [1.12.18]$$

If the values of η_T or η_L are not specified, they are derived from Equation 1.12.19.

$$\eta_T = -\frac{1}{\tan(2\alpha_0)}, \eta_L = \frac{S_{lX} \cdot \cos(2\alpha_0)}{Y_{lC} \cdot \cos(\alpha_0)^2} \quad [1.12.19]$$

The parameter α_0 represents the fracture plane angle for pure transverse compression. Several sources report that $51° \leq \alpha_0 \leq 55°$ is typical for glass and carbon composites [5] [8] [9]. A default value of 53 degrees is assumed.

If the value of φ^0 is not specified then it can be derived iteratively from Equations 1.12.20-21, otherwise a value of 0 degrees is assumed.

$$\varphi^0 = \varphi^c - \gamma_{m^0} \left[\frac{1}{2} \sin(2\varphi^0) X_{lC} \right] \quad [1.12.20]$$

$$\varphi^c = \tan^{-1} \left[\frac{1 - \sqrt{1 - 4\left(\frac{S_{lX}}{X_{lC}} + \eta_L\right)\frac{S_{lX}}{X_{lC}}}}{2\left(\frac{S_{lX}}{X_{lC}} + \eta_L\right)} \right], 4\left(\frac{S_{lX}}{X_{lC}} + \eta_L\right)\frac{S_{lX}}{X_{lC}} \leq 1 \quad [1.12.21]$$

You specify the iterative calculation method for φ^0 in the material definition[1].

The iterative calculation may be time-consuming if the assessment is run for multiple analysis items. The user must ensure that the inequality in Equation 1.12.21 is satisfied in the material data.

Critical plane searching

The values of α and ψ which maximise the functions $F_{I,m}$ and $F_{I,k/s}$ are found by performing critical plane analyses for $0 \leq \alpha, \psi \leq 180$, respectively.

[1] Requires the Symbolic Math Toolbox.

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